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Read my Mind

EEG Motor Imagery Decoding with the Multi-Scale Dual-Axis
Conformer

Walter J. Troiani Josu Bernal

January 26, 2026

1. Introduction & Context
2. Preprocessing & Augmentation
3. Learning architectures
4. Results
5. Conclusions

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Motivation

BCI-MI offers a non-invasive way to translate neural intentions. However, the practical deployment often leads to poor generalization across different users or sessions and the development of a unified framework remains an open challenge.

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BCI-MI offers a non-invasive way to translate neural intentions. However, the practical deployment often leads to poor generalization across different users or sessions and the development of a unified framework remains an open challenge.

Goals

- Standardized Preprocessing and Augmentation
- Systematic Literature Review
- Novel Architecture and Validation

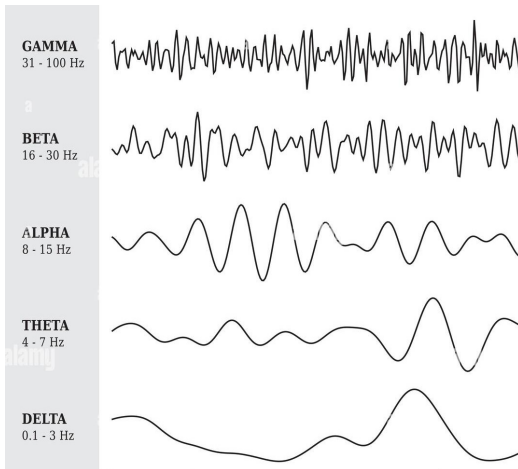
Largest EEG BCI dataset publicly available (60h)

- Three distinct experiments, each with increasing levels of complexity: CLA, HaLT and 5F.
- MATLAB format
- Two-dimensional array with dimensions $nS \times 22$ ($19 + 2 + 1$),
- 200 Hz (1000 Hz)
- 10-20 system (standard placing)

Understanding Electroencephalogram Data



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1. Complex, non-linear dynamics
2. Artifacts (e.g. heartbeats, eye blinks)
3. Biological expertise
4. Generalization across different users or sessions

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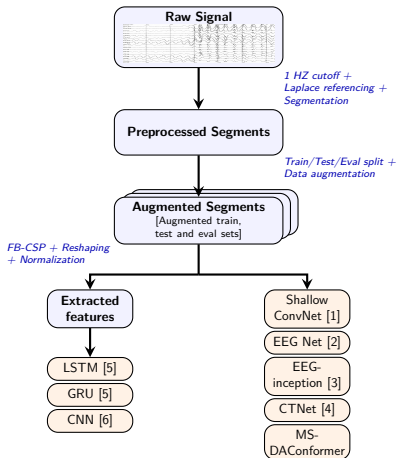
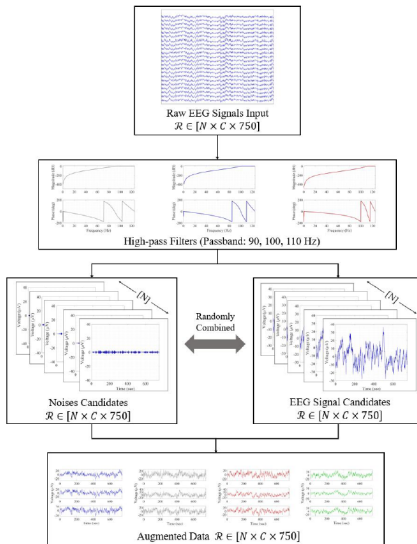


Figure: High level pipeline diagram

Data Augmentation



Nyquist-Shannon Sampling Theorem

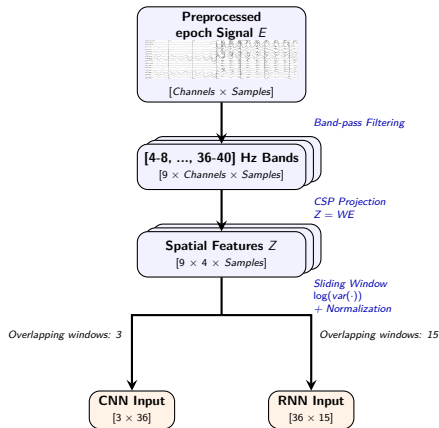


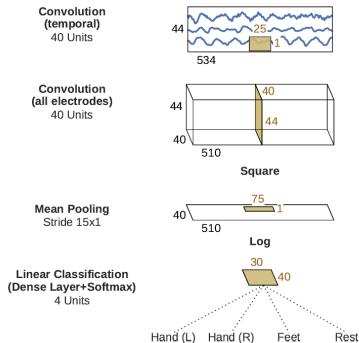
Figure: Feature extraction diagram

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1. CNN
2. RNN
 - LSTM
 - GRU

Shallow ConvNet (2017) [1]

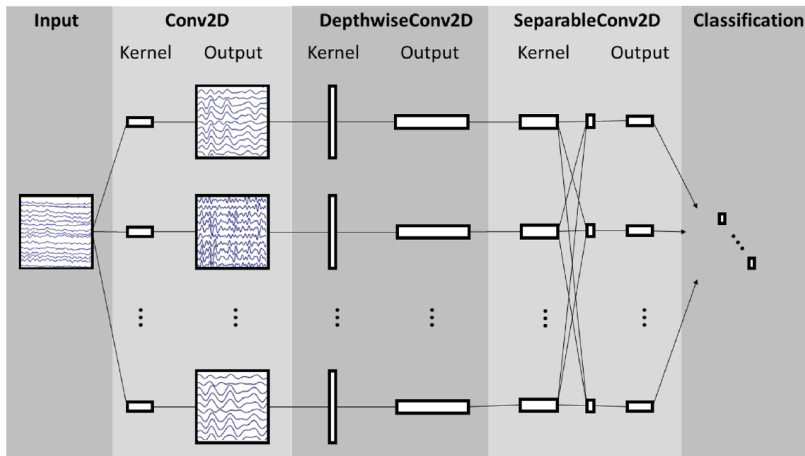


1. Designed for oscillatory signal classification
2. Inspired by FBCSP
 - The first two layers perform a temporal convolution and a spatial filter
 - Squaring nonlinearity, a mean pooling layer, and a logarithmic activation function

General framework, limited data, and interpretable.

1. **Block 1:** Two sequential convolutional steps (FB-CSP)
 - 1.1 Kernel at **half the sampling rate**
 - 1.2 Depthwise convolution
2. **Block 2:** Separable Convolution (depthwise + pointwise)

EEG Net (2018) [2]



Inception module ensemble,
adapted from computer vision.

- Depth (series) and width (parallel) features.
- Multi-scale processing captures diverse temporal patterns.

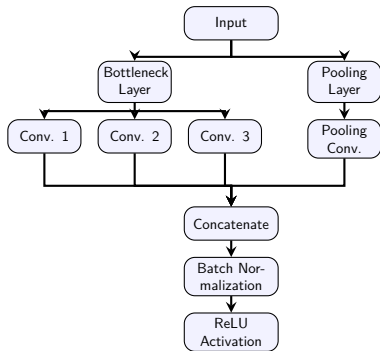
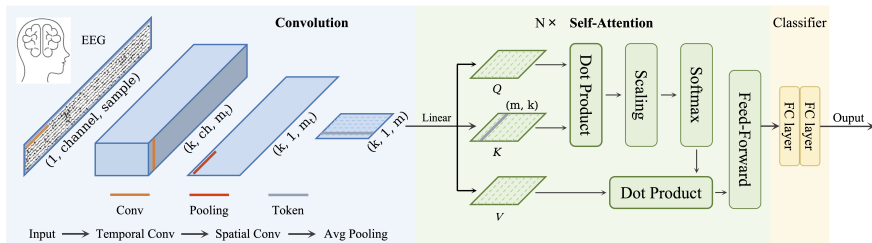


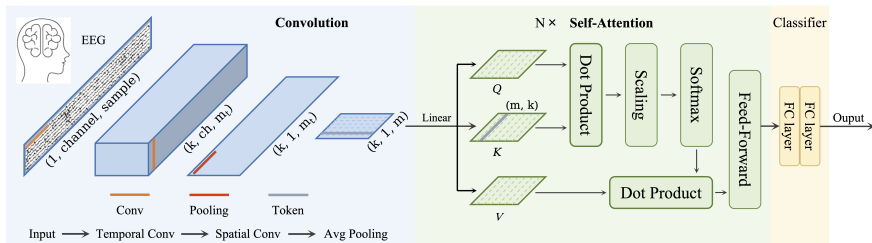
Figure: Inception module diagram.

EEG-Conformer (2022) [7]



EEG-Conformer architectures typically consist of:

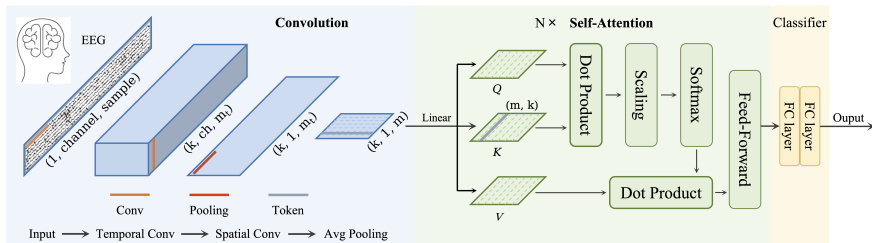
EEG-Conformer (2022) [7]



EEG-Conformer architectures typically consist of:

1. **Convolutional patch embedding** front-end (ShallowConvNet) for local feature extraction

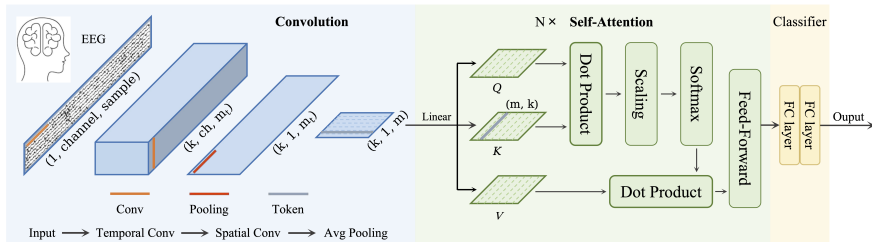
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1. **Convolutional patch embedding** front-end (ShallowConvNet) for local feature extraction
2. **Self-attention encoders** across temporal and spatial (channel) dimensions
3. **MLP classifier** for final prediction



- **EEGNet-style Patch Embedding** Use of Depthwise convolutions preserve channel structure and avoid early spatial compression

CTNet: Convolutional Transformer Network (2024) [4]



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- **Positional encoding** Explicit temporal ordering for EEG temporal tokens

CTNet: Convolutional Transformer Network (2024) [4]



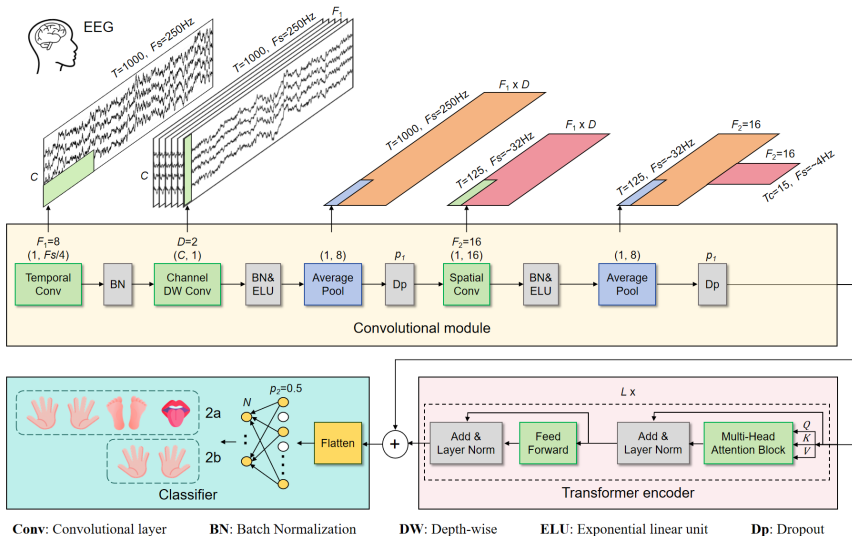
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- **EEGNet-style Patch Embedding** Use of Depthwise convolutions preserve channel structure and avoid early spatial compression
- **Positional encoding** Explicit temporal ordering for EEG temporal tokens
- **Residual connections** Patch embeddings are skip-connected to attention outputs to enhance regularization and representation (ResNet)

CTNet: Convolutional Transformer Network (2024) [4]



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Multi-Scale Dual-Axis Conformer (2026) [8]



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CTNet and EEG-Conformer have strong inductive biases:

- **Temporal first embeddings:** dependencies over time are prioritized rather than channels.
- **Single-scale local patterns:** fixed convolutional kernel (long) hopefully capture all frequency bands at one temporal scale.

How could we generalize it?

Multi-Scale Dual-Axis Conformer (2026) [8]



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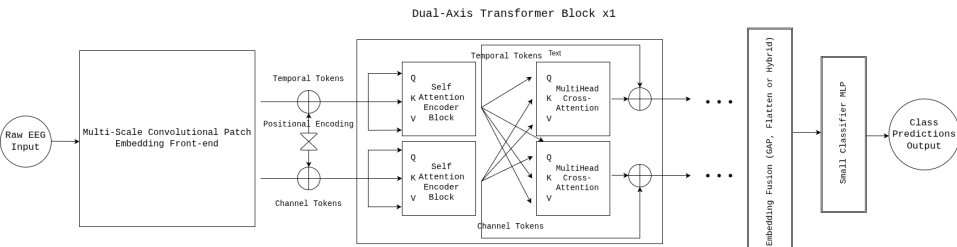
How could we generalize it?

- **Multi-scale temporal features:** promote the capture of diverse EEG rhythms ($\alpha, \beta, \gamma, \delta, \theta$) with multiple kernel lengths
- **Dual-axis interactions:** model temporal and spatial dependencies flexibly via Channel Tokens and bidirectional cross-attention

Multi-Scale Dual-Axis Conformer (2026) [8]



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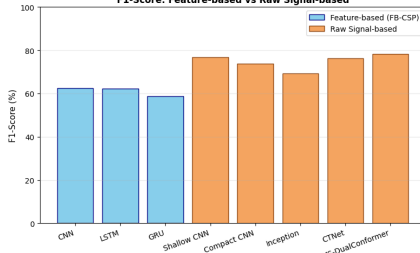
Results 1: CLA (3 classes)



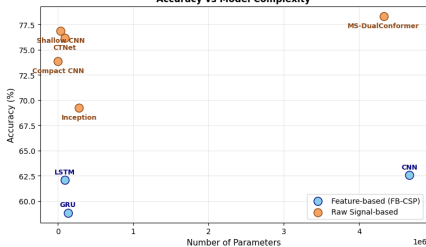
Test Accuracy: Feature-based vs Raw Signal-based



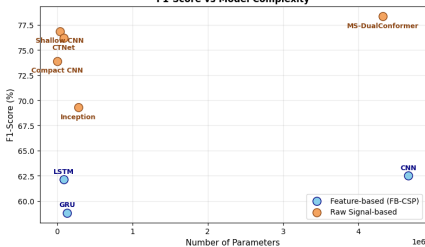
F1-Score: Feature-based vs Raw Signal-based



Accuracy vs Model Complexity



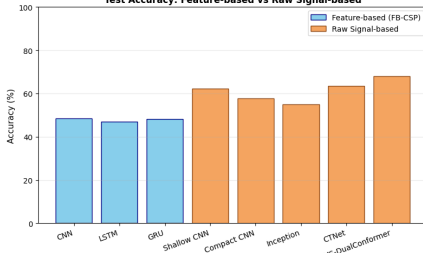
F1-Score vs Model Complexity



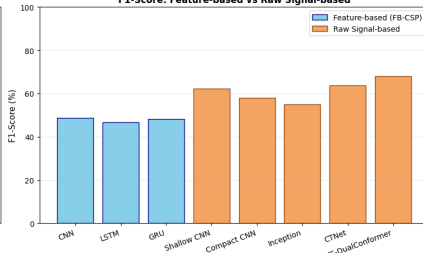
Results 2: HaLT (6 classes)



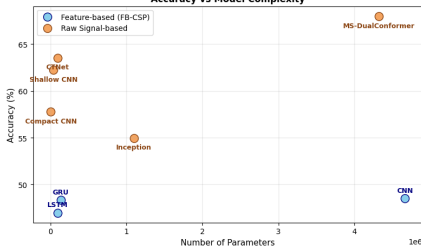
Test Accuracy: Feature-based vs Raw Signal-based



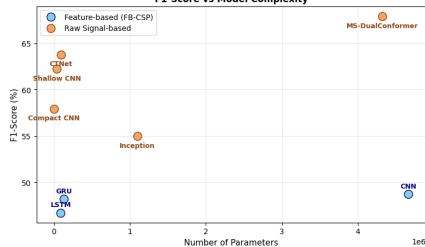
F1-Score: Feature-based vs Raw Signal-based



Accuracy vs Model Complexity



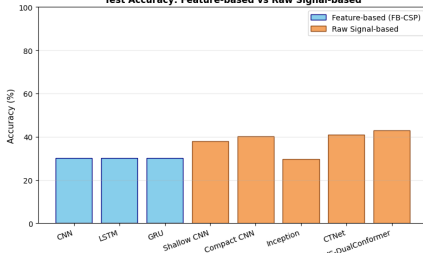
F1-Score vs Model Complexity



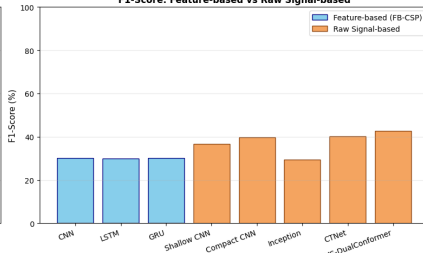
Results 3: 5F (5 classes)



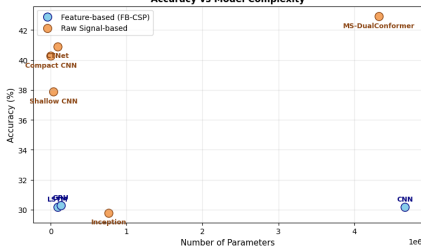
Test Accuracy: Feature-based vs Raw Signal-based



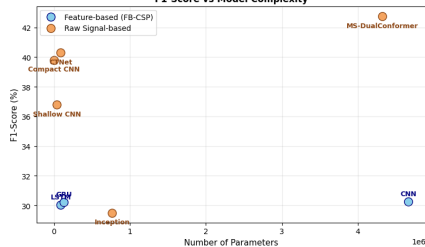
F1-Score: Feature-based vs Raw Signal-based



Accuracy vs Model Complexity



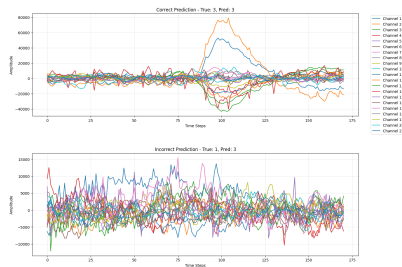
F1-Score vs Model Complexity



Demo: Raw EEG Examples



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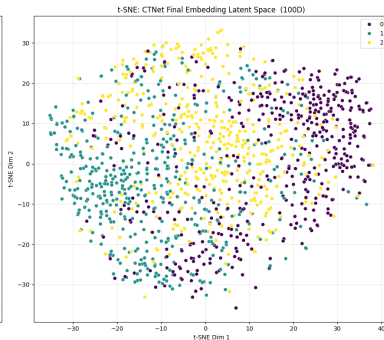
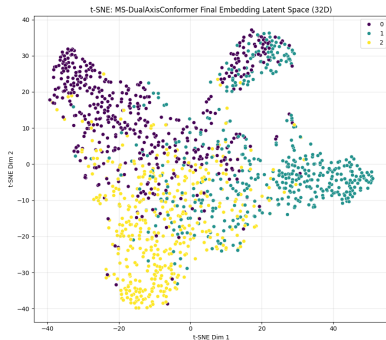


Example 1

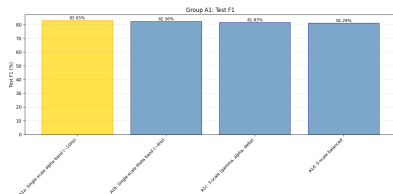


Example 2

Demo: t-SNE EEG Embeddings

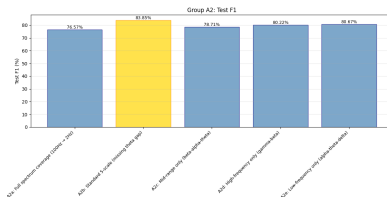


Comparison of final learned embeddings for CLA: MS-DA-Conformer vs CTNet



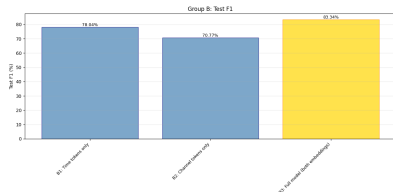
A1: Single-scale vs Multi-scale

Comparing single-scale kernels vs
limited multi-scale coverage



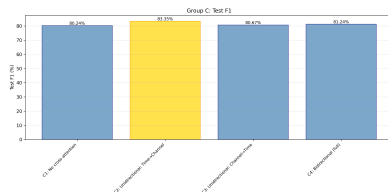
A2: Frequency-aligned Multi-scale

Evaluating neurophysiology-aligned
kernel sizes across full spectrum



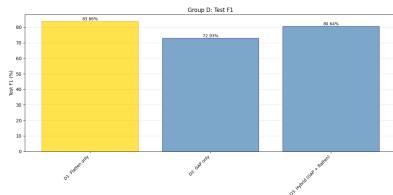
B: Temporal vs Channel Embeddings

Dual-axis embeddings outperform single-axis variants



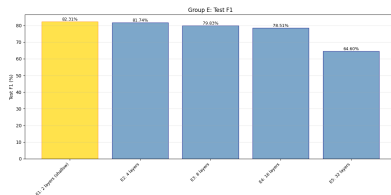
C: Cross-Attention Fusion

Time→Channel most relevant; bidirectional slightly better than none



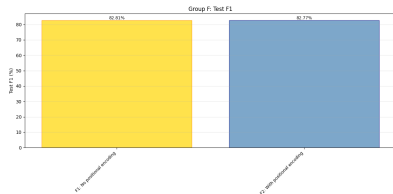
D: Pooling Strategy

Flattening fused tokens slightly outperforms GAP-only



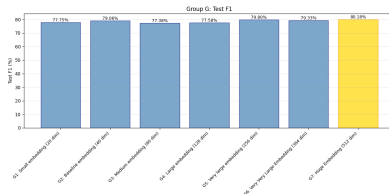
E: Attention Layer Depth

Deeper layers show diminishing returns without more data



F: Positional Encoding

Minimal impact on performance



G: Embedding Dimensionality

Small-to-mid embedding sizes
(20-64) are optimal; larger sizes
have minor gains

- **Dual-axis embeddings** (temporal + channel) its better than temporal only.
- **Cross-attention fusion** improves representation;
Time→Channel is most relevant
- **Flatten pooling** far better than GAP although tradeoff.
- **Multi-scale embeddings** do not consistently outperform single-scale
- **Positional Encoding**: It does not seem to have any influence
- **Embedding Dimensionality**: Larger embeddings are not better in this case.

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Key Findings

The MSDAC outperforms SOTA architectures by explicitly modeling both temporal dynamics and spatial dependencies.

- Dual embeddings with temporal→channel cross-attention provide a clear architectural advantage.
- Multi-scale patch embeddings do not yield consistent performance gains.
- Embedding sizes in the range of 20–64 are optimal; positional encoding has minimal impact.

Despite its performance, architectural complexity can be pruned. Further improvements will require deeper neurological **domain knowledge** and **EEG signal inductive biases**.

Future Directions

Future research should focus on:

- **Revisiting EEG-Conformer** to fix instability and include in benchmarks.
- **Optimizing the Pareto frontier** by simplifying multi-scale components, refining embedding dimensionality, and possibly restricting cross-attention.
- Enhancing generalization for challenging tasks like subtle finger-movement classification (**5F task**).
- **Exploring novel architectures beyond the Conformer** paradigm, such as GNN-based Transformers or Convolutional Spiking Neural Networks.

Thank you so much for your **Attention!**



Any questions?



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